Here’s a step-by-step guide to create a simple web application that demonstrates route grouping in Laravel 12.

4.10.4.1 Step 1: Create and Open Laravel Project

We will create a new Laravel project named `lab-group` using Composer. Open your terminal and run:

```
composer create-project laravel/laravel:^12.0.3 lab-group
cd lab-group
code .
```

You should see the Laravel project structure in Visual Studio Code.

4.10.4.2 Step 2: Create a Controller

```
php artisan make:controller PageController
```

Update the controller `app/Http/Controllers/PageController.php`:

```
namespace App\Http\Controllers;

class PageController extends Controller
{
    public function home()
    {
        $message = "Welcome to the homepage.";
        return view('pages.home', compact('message'));
    }

    public function about()
    {
        $message = "This is the about page.";
        return view('pages.about', compact('message'));
    }

    public function contact()
    {
        $message = "Reach us through the contact page.";
        return view('pages.contact', compact('message'));
    }
}
```

Explain the code:

- The `PageController` is created in the `app/Http/Controllers` directory.
- The `PageController` has three methods: `home()`, `about()`, and `contact()`.
- Each method returns a view with a message passed to it.